Data Quality Report — BOAMP Renewal Modeling Dataset

Gigalis Internship — Phase 1 Deliverable L1

Internship Project — Gigalis

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1 Objective and Scope

This report documents the quality of the BOAMP corpus used to construct the renewal proxy and the survival modeling dataset. The scope is the BOAMP-only pipeline for digital-sector public procurement notices in Pays de la Loire over 2015–2024. The official modeling population is the set of eligible `APPEL_OFFRE` contracts retained in `data/processed/boamp_phase2_survival.csv`. In this report, `event = 1` denotes a *proxy renewal event* found by the linking algorithm; `event = 0` denotes right-censoring or no observed link under the rule before the study end date. It does not prove that no renewal occurred.

2 Raw Corpus Overview

The cleaned BOAMP corpus is stored in `data/processed/boamp_full_clean.csv`. It contains 3,181 notices published between 2015-03-02 and 2024-12-29.

Table 1: Raw BOAMP notice types

Notice type	Count	Share
APPEL_OFFRE	1,933	60.8%
ATTRIBUTION	1,081	34.0%
RECTIFICATIF	150	4.7%
PRE-INFORMATION	7	0.2%
MODIFICATION	5	0.2%
INTENTION_CONCLURE	4	0.1%
PERIODIQUE	1	0.0%
Total	3,181	100.0%

The renewal-linking pipeline uses only the 1,933 `APPEL_OFFRE` notices as possible source contracts. `ATTRIBUTION` notices are used as an auxiliary source to refine contract start dates when a BOAMP back-reference is available.

3 Final Modeling Population

The study population is narrower than the raw corpus because not every notice is a candidate source contract and not every source contract has an observable renewal window before the 2024-12-31 study end.

Table 2: Population construction summary

Population	Count	Interpretation
Raw BOAMP notices	3,181	Full digital-sector corpus downloaded from the BOAMP API.
APPEL_OFFRE source notices	1,933	Potential source contracts for renewal linking.
Eligible source contracts	1,100	APPEL_OFFRE contracts whose estimated end date falls before study end.
Linked proxy-renewal events	697	`event = 1`; a plausible later BOAMP notice was linked.
Right-censored contracts	403	`event = 0`; no observed link under the rule before study end.

The linking rate in the eligible population is 63.36% (697/1,100). This is the denominator that matters for Phase 2 survival modeling. The lexical baseline retained in the repository for comparison is 11.3%.

4 Missing-Value and Cleaning Summary

Table 3 summarizes the main variable-level quality issues that carry forward into modeling.

Table 3: Variable quality and cleaning decisions

Variable	Issue	Cleaning / treatment	Modeling impact
`buyer_siret` / `buyer_key`	Raw `buyer_siret` coverage is 9.1% in the 3,181-row corpus; all 1,100 eligible contracts use a name-based `buyer_key`.	Build `buyer_key` as `SIRET:...` when valid, else `NAME:...`; no manual buyer reconciliation.	Buyer-name fragmentation is the main source of false negatives in linking.
`CPV` fields	`cpv_principal` is missing for 1.9% of the raw corpus; generic codes reduce specificity. Eligible `cpv_div2` coverage is 95.3%.	Clean and standardize CPV; generic codes are capped at division-level similarity; missing CPV is excluded from the composite score with weight renormalization.	Missing or generic CPV weakens auditability and may reduce match discrimination.
`amount_clean`	Valid amount coverage is 41.1% in the full corpus and 21.4% among `APPEL_OFFRE`; 20.3% in the eligible handoff.	Flag zero, tiny, and ceiling values; set flagged amounts to missing.	Amount is too sparse to act as a linking signal and must be handled as optional context only.
`declared_duration_months` `duration_clean`	Raw cleaned duration coverage is 47.3% overall and 76.6% for `APPEL_OFFRE`. In the eligible cohort, 25.1% are imputed.	Remove suspect durations and impute missing eligible durations to 48 months with `dur_was_imputed`.	Duration affects eligibility and temporal search windows, so imputed rows are less reliable.
`start_date`	No missing values in the eligible handoff, but 59.3% use publication date and 40.7% use an attribution-derived award date.	Use award date via `annonce_lie` when available, else publication date.	Temporal scores contain date noise because publication is not always the operational contract start.
`type_procedure`	0.3% missing in the eligible handoff.	Retain as-is; no imputation required.	Low risk for descriptive summaries and covariate use.
`type_marche`	14.4% missing in the eligible handoff.	Retain missing values explicitly.	Missingness may blur segment-level interpretation.
`event` / renewal link	No explicit BOAMP renewal identifier exists; `event` is algorithmic rather than observed ground truth.	Use the linking algorithm to build a proxy event; keep confidence diagnostics.	False positives and false negatives remain possible, so event quality must be interpreted as bounded evidence.

5 Key Data Quality Issues

The dominant data quality problems are structural rather than cosmetic. Buyer identity is weak because the eligible cohort has no SIRET-anchored `buyer_key`; most grouping therefore relies on normalized buyer names. Enriching buyer identification with SIRET/SIREN data is a planned future improvement that requires a cross-source join unavailable in the current BOAMP-only pipeline. CPV coverage is high but not perfect, and generic CPV values reduce matching precision. Contract amount is sparse on source notices, so it cannot support renewal linking. BOAMP also has no explicit renewal label, which means the linking output is necessarily an inferred event proxy.

6 Duration Field Reliability

Declared duration is not the survival target. In the `APPEL_OFFRE` population, the cleaned declared duration field is heavily concentrated on administrative values of 12, 24, 36, and 48 months.

Table 4: Declared duration concentration in APPEL_OFFRE notices

Duration value	Count	Share of non-missing durations
48 months	462	31.2%
12 months	316	21.4%
24 months	151	10.2%
36 months	99	6.7%
Subtotal exactly 12/24/36/48	1,028	69.5%

This concentration is more consistent with administrative ceilings than with realized renewal timing. For that reason, `declared_duration_months` is used as a covariate and as a temporal search prior, while `observed_duration_months` is the survival time used in Phase 2.

7 Renewal Linking Quality

BOAMP provides no explicit renewal ID, so the renewal variable is constructed by matching later notices from the same `buyer_key` using text, CPV, and timing. The eligible modeling dataset contains 697 linked proxy-renewal events and 403 right-censored contracts.

Table 5: Linking outcome summary

Metric	Value	Interpretation
Linked events	697	Proxy renewal events retained in the official BOAMP hand-off.
Right-censored	403	No observed link before study end under the chosen rule.
Linking rate	63.36%	Share of eligible contracts with `event = 1`.
Strict high-confidence links	90	Conservative subset with composite score ≥ 0.70 and score margin $\geq$ 0.05, *excluding* the 135 single-candidate links (margin undefined). A broader count of 104 results when single-candidate links with score ≥ 0.70 are also included.
Single-candidate links	135	Linked contracts with only one surviving candidate after filtering; margin is undefined for these.
Ambiguous multi-candidate links	273 of 562	48.58% of multi-candidate events have a score margin below 0.05.
Main unlinked failure mode	339 of 403	Most unlinked contracts have no temporal partner in the search window.
Secondary failure modes	32 text, 32 CPV	Some eligible contracts had a temporal partner but failed semantic or CPV checks.

The confidence diagnostics are useful for sensitivity analysis, not as legal ground truth. A strict subset can bound the event definition, but it does not prove that the remaining links are false.

8 Biases and Limitations

Selection bias remains possible because contracts with no temporal partner cannot be linked even if a real-world renewal occurred outside the observable window. Buyer fragmentation can create false negatives, while generic or repetitive contract descriptions can create false positives. Missing or generic CPV reduces auditability. Contracts from late 2023 face compressed observation windows, and 2024 contracts are structurally excluded from the eligible denominator because their expected renewal window falls after study end. Manual validation is still required to estimate the precision of the proxy renewal event.

Table 6: Principal data quality risks

Risk	Evidence in current data	Modeling impact
Buyer fragmentation	All 1,100 eligible rows use name-based `buyer_key` values; no SIRET-anchored keys appear in the eligible cohort. SIRET/SIREN enrichment is a planned future improvement.	Structural false negatives when the same buyer appears under multiple spellings.
Generic or missing CPV	Raw CPV missingness is 1.9%; generic codes are capped at low similarity.	Lower match precision and weaker audit trail for some links.
Sparse amount field	Only 21.4% of source `APPEL_OFFRE` notices have a usable amount.	Amount cannot serve as a reliable matching signal or segmentation variable.
Administrative duration values	69.5% of observed source durations equal 12, 24, 36, or 48 months.	Declared duration should not be interpreted as realized renewal time.
Observation-window censoring	403 eligible contracts are right-censored; late cohorts have less time to reveal renewals.	Apparent non-renewal may reflect study-end truncation rather than true absence of renewal.
Proxy event definition	BOAMP contains no explicit renewal label.	Event quality depends on the linking rule and should be stress-tested with confidence diagnostics.

9 Reproducibility

This report is reproducible from tracked project outputs. The core files are:

- `data/processed/boamp_full_clean.csv`
- `data/processed/boamp_phase2_survival.csv`
- `boamp_renewal_linking_quality/outputs/boamp_linking_stats.csv`
- `boamp_renewal_linking_quality/outputs/boamp_renewal_links.csv`
- `boamp_renewal_linking_quality/outputs/boamp_renewal_candidates.csv`
- `boamp_renewal_linking_quality/outputs/boamp_bias_report.csv`
- `boamp_renewal_linking_quality/outputs/boamp_link_confidence_diagnostics.csv`

The relevant pipeline steps are implemented in `scripts/task_boamp_full_clean.py`, `scripts/task9_boamp_phase2_handoff.py`, and `scripts/task10_link_confidence_diagnostics.py`. No manual, untracked data manipulation is required to reproduce the counts reported here.

10 Conclusion

The BOAMP dataset satisfies the internship guide's minimum data-quality requirement for Phase 1 because it provides a documented corpus, explicit cleaning rules, missingness information, a transparent linking rate, and a survival-ready population of more than 800 eligible contracts with either observed or censored duration. The main limitation is conceptual rather than technical: `event` remains a proxy renewal event, not legal ground truth, so later manual validation is still needed to estimate precision.